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Canonical STATIS: Biplot analysis of multi-table group structured data based on STATIS-ACT methodology

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Abstract

Frequently multivariate experimental data taken over multiple occasions is found to produce a multivariate multi-table experiment. Neither the separate analysis of each occasion, using MANOVA or canonical variate analysis, nor the joint analysis using the STATIS-ACT methodology for multiple tables, are adequate to capture the real structure of the data matrices, since the first account for group structure but not time evolution and the second confuses the "between" and "within" group variabilities. A method baptized Canonical STATIS, based on STATIS-ACT methodology, is proposed. The method permits to account for data group structure as well as time evolution on the various occasions by obtaining common or stable canonical variates across multiple occasions or data sets. A simultaneous representation of groups, variables and occasions (biplot) is obtained, thereby widening the capabilities of classical methods.
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1. Introduction

Due to recent advances in data collecting, it is every time more frequent to find experimental situations in which one can see measurements of several variables over a set of individuals classified a priori in groups, and repeated across multiple occasions or conditions. This type of data is produced, for example, when the groups correspond to various treatments in a multivariate experimental design, at successive occasions. These could be called multi-table multivariate data with group structure. In the literature, this kind of data is often referred as doubly multivariate repeated measures data.

If we denote by K the number of occasions or conditions, the data are contained in matrices $\mathbf{X}_k$ $(k = 1, 2, \ldots, K)$, of the size $(I \times J)$, where J is the number of variables and I the number of individuals, who are classified in G mutually exclusive groups of the size I_g $(g = 1, \ldots, G)$ with $\sum_g I_g = I$. Without loss of generality, we can assume that every matrix is column centered. Let $\overline{\mathbf{X}}_k$ be the matrix containing the means of each group for each variable (when the data matrix is column centered contains the effects of the treatments), $\mathbf{D} = diag(I_1, \ldots, I_G)$ the diagonal

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matrix containing the group sample sizes, $\mathbf{B}_k = \left(\overline{\mathbf{X}}_k^{\mathrm{T}}\mathbf{D}\overline{\mathbf{X}}_k\right)/(G-I)$ the "between-groups" covariance matrices and $\mathbf{S}_k = \left(\mathbf{X}_k^{\mathrm{T}}\mathbf{X}_k - \overline{\mathbf{X}}_k^{\mathrm{T}}\mathbf{D}\overline{\mathbf{X}}_k\right)/(I-G)$ the pooled within groups covariance matrices for each occasion.

Observe that, although here the method is described for the same variables measured in the same individuals at each occasion, the method would be valid for different sets of variables at each occasion or even different sets of variables at one occasion.

The classical approach to analyze this type of data is to use multivariate analysis of variance (MANOVA) for every occasion, or Canonical variate analysis (CVA) from the geometric point of view, Gittins (1985). For a recent description see, Gardner et al. (2006). CVA allows us to derive linear combinations that successively maximize the ratio of "between-groups" to "pooled within-group" sample variance, obtained from the eigenvectors of $\mathbf{S}_k^{-1}\mathbf{B}_k$, and with maximum discriminatory power.

In its biplot form (Gabriel, 1995; Gower and Hand, 1996; Amaro et al., 2003; Varas et al., 2005) one obtains joint representations of the group means and variables in which it is possible not only to establish the differences between groups but also to characterise the variables responsible for them. The biplot represents the matrix $\overline{\mathbf{X}}_k$ of the means of the treatments/groups, using adequate metrics ($\mathbf{D}$ and $\mathbf{S}_k^{-1}$ for rows and columns, respectively). More precisely, the biplot $\overline{\mathbf{X}}_k = \mathbf{A}_k\mathbf{B}_k^{\mathrm{T}}$ is obtained from the ordinary singular value decomposition of $\mathbf{D}^{1/2}\overline{\mathbf{X}}_k\mathbf{S}_k^{-1/2} = \mathbf{P}_k\Delta_k\mathbf{Q}_k^{\mathrm{T}}$, or equivalently from the generalized singular value decomposition $\overline{\mathbf{X}}_k = \mathbf{D}^{-1/2}\mathbf{P}_k\Delta_k\mathbf{Q}_k^{\mathrm{T}}\mathbf{S}_k^{1/2} = \mathbf{U}_k\Delta_k\mathbf{V}_k^{\mathrm{T}}$.

$\mathbf{U}_k$ provides a reference system in $\mathbb{R}^G$ in which $\overline{\mathbf{X}}_k^{\mathrm{T}}$ can be projected to obtain its principal components. In the same way $\mathbf{V}_k$ provides a reference system in $\mathbb{R}^J$ in which $\overline{\mathbf{X}}_k$ can be projected. The markers for rows and columns are the first columns of

$$\mathbf{A}_k = \mathbf{U}_k\Delta_k = \mathbf{D}^{-1/2}\mathbf{P}_k\Delta_k, \quad \mathbf{B}_k = \mathbf{V}_k = \mathbf{S}_k^{1/2}\mathbf{Q}_k. \tag{1}$$

In this way, the means are represented in a subspace of low dimension, with reference axis of maximum discrimination between groups, and with optimal quality of representation. The Euclidean distance between group means approximates the Mahalanobis distance, thus avoiding redundant information. These solutions give information about data structure in each occasion but, in general, it is not simple to analyze simultaneously the similarities and the differences between occasions.

When one tries to analyze the evolution of distinct structures during a period of time, extracting its common features, the STATIS-ACT method (Escoufier, 1980; L'Hermier des Plantes, 1976; Lavit et al., 1994) provide an alternative to analyze which is less known than the above but is being more and more used. STATIS-ACT considers the scalar products $\mathbf{W}_k = \mathbf{X}_k\mathbf{X}_k^{\mathrm{T}}$ as a representative object of each occasion, defines a measure of similarity between occasions and search for the object most similar to all the objects as a weighted mean $\mathbf{W}_c = \sum_k \alpha_k\mathbf{W}_k$. The underlying true consensus is then estimated by means of the principal components of the weighted mean.

This solution does not capture the structure of data matrices because it works with the total variability thus confounding the between and within groups variabilities. The alternative of considering the STATIS method on the matrices of means does not solve the problem either, because ignores the within groups variability, which is necessary to establish the significance of differences. Other three-way methods or multi-table methods as *three-mode principal component analysis* (Kroonenberg, 1983), or the *general euclidean model* (Young, 1984, 1987) could have been used. None of the alternative models take into account the group structure or the generalized metrics. Adding the group structure and the generalized metrics would be the subject of another research. Taking into account the close relationship (Husson and Pagès, 2006) between the RV coefficient, used in STATIS, and the INDSCAL model, a part of the general Euclidean model, the extension is probably straightforward.

In the present work we propose an alternative that obtains a common reference subspace for the canonical analysis of all the occasions. It is specifically designed to capture the stable structure of the differences on the group means over time. The projections of all the data matrices onto this common reference subspace (obtaining a biplot representation) allows for the visualization of the similarities and differences over time and the characterization of the variables responsible for them.

The analysis can be carried out from two different points of view: searching for a common structure on Mahalanobis distances among means, or searching for a common structure on covariances among variables. In this work we take first of these two points of view, leading us to a technique baptized canonical STATIS and the corresponding biplot representations.

It should be pointed out that the main mathematical results concerning STATIS-ACT, including metrics different from the identity, are completely established by its original authors, although Mahalanobis metric was not specifically used and no applications with generalized metrics have been found. The main contribution of this work is looking at the compromise as a way of obtaining a common reference system for all occasions allowing for the definition of a biplot representation, new for these kind of techniques. As it is well known, biplot allows for the simultaneous representations of the rows and the columns of a data matrix while STATIS-ACT obtains a graphical representation for the rows only.

2. The method

Canonical STATIS has the same structure as classical STATIS-ACT except that it uses the Mahalanobis metric instead of the usual Euclidean metric to measure differences among group means. First, we obtain a graphical representation of the occasions, based on a measure of similarity among them (inter-structure), then construct a reference system common to all occasions (search for compromise) and finally, projecting the original data on the compromise, we obtain a biplot representation on a common reference subspace.

2.1. Definitions

A *study* is defined as a triplet composed by the mean matrix and the row and column metrics. That is $\left(\overline{\mathbf{X}}_k, \mathbf{S}_k^{-1}, \mathbf{D}\right)$ with $k = 1, 2, \ldots, K$. As a metric for measuring differences among mean vectors we use Mahalanobis' (the inverse of the within groups covariance matrix $\mathbf{S}_k^{-1}$), and the metric for variables we use $\mathbf{D} = diag\,(I_1, \ldots, I_G)$ the diagonal matrix containing the group/treatments sample sizes.

As a representative *object* for the *study* we take scalar product matrix $\mathbf{W}_k = \overline{\mathbf{X}}_k \mathbf{S}_k^{-1} \overline{\mathbf{X}}_k^{\mathrm{T}}$, which defines Mahalanobis scalar products/distances among means, i.e., comparison among studies is carried out with the structure of Mahalanobis inter-distances among means in the multidimensional generalized space. The principal coordinates/components analysis of each of the objects, with respect to the metric $\mathbf{D}$, is closely related to CVA in the sense that it contains representation of the means on the canonical subspace with maximum discriminatory power. (See, Gabriel, 1995.)

We define the *Hilbert–Schmidt scalar product* between two objects $\mathbf{W}_k$ and $\mathbf{W}_{k'}$ for $k, k' = 1, \ldots, K$, as

$$\langle \mathbf{W}_k | \mathbf{W}_{k'} \rangle_{HS} = Tr\left(\mathbf{D}\overline{\mathbf{X}}_k \mathbf{S}_k^{-1} \overline{\mathbf{X}}_k^{\mathrm{T}} \mathbf{D} \overline{\mathbf{X}}_{k'} \mathbf{S}_k^{-1} \overline{\mathbf{X}}_{k'}^{\mathrm{T}}\right).$$

The scalar product of representative objects leads us to establish a measure of the proximity between two studies. We denote by $\mathbf{S}^W$ the matrix that contains scalar products for each pair of occasions. The diagonal of this matrix contains the squared norm of all objects; when the norms of each study are very different, it is convenient to normalize dividing each object by its norm. If studies are normalized, then scalar products coincide with the *RV* correlation coefficient (Escoufier, 1973):

$$RV\,(\mathbf{W}_k, \mathbf{W}_{k'}) = \frac{\langle \mathbf{W}_k | \mathbf{W}_{k'} \rangle}{\sqrt{\langle \mathbf{W}_k | \mathbf{W}_k \rangle \langle \mathbf{W}_{k'} \mathbf{W}_{k'} \rangle}}.$$

Recent uses of the *RV* coefficient can be found in Husson and Pagès (2006).

In general we will use normalized studies. The procedure is organized, as in STATIS-ACT, with three successive stages: inter-structure, compromise and intra-structure.

2.2. Inter-structure: global similarities between occasions

The inter-structure analysis has its objective to decide whether or not there is a common structure to all K representative objects $\mathbf{W}_k$. The analysis of similarities among objects is obtained from representing principal coordinates/components of the matrix $\mathbf{S}^W$. The coordinates $\mathbf{E}$ of the points are obtained from the spectral decomposition $\mathbf{S}^W = \mathbf{L}\Gamma\mathbf{L}^{\mathrm{T}}$, taking $\mathbf{E} = L\Gamma^{1/2}$.

This analysis provides a graphical representation in which each study is represented by a vector; if the vectors have similar lengths and the angles between them are small, we can conclude that a common structure exists. (See Fig. 1 in the application).

2.3. Compromise: searching for a common reference system

To search for a common reference system to all the studies, we construct a compromise object denoted by $\mathbf{W}_c$, of the same nature of all the objects $\mathbf{W}_k$, which is a good resume of all. We take this object as $\mathbf{W}_c$, the object most correlated, in the sense of Hilbert–Schmidt scalar product, to all the objects $\mathbf{W}_k$. Lavit (1988) proved that the desired object is the weighted mean of the objects $\mathbf{W}_k$, in the form

$$\mathbf{W}_c = \sum_{k=1}^{K} \alpha_k \mathbf{W}_k. \tag{2}$$

The coefficients α_k are determined by

$$\alpha_k = \frac{1}{\sqrt{\gamma_1}} \left(\sum_{\tau=1}^{K} \sqrt{s_{kk}} \right) l_{k1},$$

where, l_{k1} is the kth component of the first eigenvector of $\mathbf{S}^W$ associated to the largest eigenvalue γ_1 and s_{kk} is the quadratic norm of the kth object. The ideal objects whose scalar products are contained in $\mathbf{W}_c$ are what we call *compromise means*. A graphical representation of the compromise is obtained from the principal components/coordinates analysis of the matrix $\mathbf{W}_c$, with respect to the metric $\mathbf{D}$, from the spectral decomposition $\mathbf{D}^{1/2}\mathbf{W}_c\mathbf{D}^{1/2} = \mathbf{P}\Delta\mathbf{P}^{\mathrm{T}}$, or also $\mathbf{W}_c = \mathbf{D}^{-1/2}\mathbf{P}\Delta\mathbf{P}^{\mathrm{T}}\mathbf{D}^{-1/2} = \mathbf{U}\Delta\mathbf{U}^{\mathrm{T}}$, where $\mathbf{U}$ are generalized eigenvectors with the metric $\mathbf{D}$, and verifying $\mathbf{U}^{\mathrm{T}}\mathbf{D}\mathbf{U} = \mathbf{I}$.

The compromise mean vectors are obtained by taking as axis the first column of the matrix:

$$\mathbf{A}^c = \mathbf{D}^{-1/2}\mathbf{P}\Delta^{1/2} = \mathbf{U}\Delta^{1/2}. \tag{3}$$

The group means are represented on the direction of maximum discriminatory power most similar to the each one of the individual studies, and could be interpreted as the discriminatory direction common to all studies.

The matrix of eigenvectors $\mathbf{U}$, obtained from the compromise object, provides a reference system, in the multidimensional space $\mathbb{R}^G$, in which all the matrices of means $\overline{\mathbf{X}}_k^{\mathrm{T}}$ can be projected to obtain a set of principal components, common to all the studies.

3. Biplot representation induced by canonical STATIS: intra-structure

In the classical STATIS-ACT methodology, trajectories are represented for each and every mean at all the occasions, projecting scalar products for each occasion on the compromise reference as

$$\mathbf{A}_k^c = \mathbf{W}_k\mathbf{D}\mathbf{U}\Delta^{1/2}. \tag{4}$$

The trajectory for a particular group is obtained by connecting positions of each mean in the gth group, for $k = 1, \ldots, K$. We may compare the position of the compromise mean with respect to each one of the individual occasions. In general, projections have low quality of representation and make no sense when observed variables are different for distinct occasions. It is for this reason that they are suppressed in the last version of the specialized software (SPAD).

We add here a biplot interpretation of the method, based on the projection of the matrices of means onto the compromise space. The biplot lead us to an approximation of the full data matrices and a trajectory for each one of the variables in every occasion. The *trajectories of the variables* are represented on the factorial subspace spanned by the *compromise means*, which we have called the *common representation space*. In this way we can investigate which variables are most important in discriminating among the groups, which have stable behavior and which interact with time. The quality of representation (goodness of fit)) is better than in the classical trajectories and the projections are well defined even when the set of variables is not the same in every occasion.

Let $\overline{\mathbf{X}}_a$, be the matrix obtained by interactively coding all the matrices $\overline{\mathbf{X}}_k$, that is, $\overline{\mathbf{X}}_a = \left[\overline{\mathbf{X}}_1 | \ldots | \overline{\mathbf{X}}_K\right]$, and let $\overline{\mathbf{Y}}_a = \left[\sqrt{\alpha_1}\overline{\mathbf{X}}_1\mathbf{S}_1^{-1/2}, \ldots, \sqrt{\alpha_K}\overline{\mathbf{X}}_K\mathbf{S}_K^{-1/2}\right]$, both of size $(I \times JK)$.

It is possible to prove that the compromise obtained in Eq. (2) coincides with the co-ordinates on the principal components of $\overline{\mathbf{Y}}_a$, with metric $\mathbf{D}$, that is to say, those obtained by spectral decomposition of $\mathbf{D}^{1/2}\overline{\mathbf{Y}}_a = \mathbf{P}\Delta^{1/2}\mathbf{Q}^{\mathrm{T}}$, or also, $\overline{\mathbf{Y}}_a = \mathbf{D}^{-1/2}\mathbf{P}\Delta^{1/2}\mathbf{Q}^{\mathrm{T}} = \mathbf{A}^c\mathbf{Q}^{\mathrm{T}}$.

Taking into account the relationship $\overline{\mathbf{X}}_a = \overline{\mathbf{Y}}_a \mathbf{D}_\alpha^{-1} \mathbf{S}_S^{1/2}$, where $\mathbf{D}_\alpha$ is a block diagonal matrix, in which each block $\mathbf{D}_{\alpha_k} = \sqrt{\alpha_k}\mathbf{I}_J$, and $\mathbf{S}_S^{1/2}$ is a block diagonal matrix in which each block is $\mathbf{S}_k^{1/2}$, is then very easy to obtain a biplot decomposition in the form $\overline{\mathbf{X}}_a = \mathbf{A}^c\mathbf{B}^{\mathrm{T}}$ by just taking $\tilde{B} = \mathbf{S}_S^{1/2}\mathbf{D}_\alpha^{-1}\mathbf{Q}$.

The matrix $\tilde{B}$ can be split into K blocks $\tilde{B}_k$ ($k = 1, \ldots, K$), one for each occasion, where $\tilde{B}_k = \overline{\mathbf{X}}_k^{\mathrm{T}}\mathbf{D}^{1/2}\mathbf{P}\Delta^{-1/2}$, i. e., the projection of mean matrices on the common reference system, in standard co-ordinates. This means that we have obtained biplot representations for each occasion

$$\overline{\mathbf{X}}_k = \mathbf{A}^c\tilde{\mathbf{B}}_k^{\mathrm{T}}, \tag{5}$$

all on the same reference system and hence directly comparable. For the representation in reduced dimension s, it is sufficient to take the first columns of $\mathbf{A}^c$ and $\tilde{B}$, thereby obtaining a lower rank approximation $\tilde{\mathbf{X}}_a$ based on the projection on the compromise structure.

The markers for rows $\mathbf{A}^c$ coincide with the co-ordinates of the compromise means given in Eq. (3) and the rows in the matrix $\tilde{\mathbf{B}}$ are the markers for the variables on all occasions. Observe that the proposed Biplot could also be used for the traditional STATIS-ACT methodology for the raw data matrices (ungrouped data) and identity metrics. That Biplot has not been described in the literature.

When the variables are the same for each occasion, a trajectory for a particular variable is obtained by connecting positions of the markers for the jth variable (jth row of the matrix $\tilde{\mathbf{B}}_k$ in the biplot defined in (5)) for $k = 1, \ldots, K$. Projections of the points forming the trajectories for a variable, onto the vector representing a treatment, show the evolution of the effect of the treatment for that variable along the different occasions.

The biplot thus obtained is of type RMP (row metric preserving), so that the mean coordinates are shown with optimal quality of representation, this means that the Mahalanobis distances between rows, rather than correlations between variables, are optimally represented. To obtain biplot of type CMP (column metric preserving), with optimum quality for variables, it suffices to multiply the co-ordinates in columns by $\Delta^{1/2}$ and those in rows by $\Delta^{-1/2}$.

Integration of various data matrices based on the subspaces generated by the principal components rather than in maximum vector correlation, can be found for instance in Martín-Rodriguez et al. (2002). The integration includes a biplot representation similar to the proposed here.

One way of measuring global quality of representation of $\tilde{\mathbf{X}}_a$, is through the use of a percentage relationship,

$$\frac{tr\left(\tilde{\mathbf{X}}_a^{\mathrm{T}}\tilde{\mathbf{X}}_a\right)}{tr\left(\overline{\mathbf{X}}_a^{\mathrm{T}}\overline{\mathbf{X}}_a\right)} \times 100,$$

which can be interpreted as total variability percentage for the data sets accounted for the first s axis.

We could also define quality of representation obtained for occasion k as

$$\frac{tr\left(\tilde{\mathbf{X}}_{a(k)}^{\mathrm{T}}\tilde{\mathbf{X}}_{a(k)}\right)}{Tr\left(\overline{\mathbf{X}}_a^{\mathrm{T}}\overline{\mathbf{X}}_a\right)} \times 100,$$

where $\tilde{\mathbf{X}}_{a(k)}$ is that part of $\tilde{\mathbf{X}}_a$ corresponding to kth table. The comparison of this measure with variability from separate canonical biplots as obtained in Eq. (1) shows how much information we have lost when working with the common structures.

Analogous reasoning makes it possible to define quality of representation for one variable in one occasion and for a full trajectory.

4. Applying to real data

The data we use in this paper belongs to a project financed by the European Union entitled *Shallow Wetland Lake Function and restoration in a changing European environment* (SWALE). In the said project a series of experiments were performed in five different countries (Sweeden, Finland, Holand, United Kingdom and Spain). For this application we have used those corresponding to Spain, where the experiment was performed around Sentiz lake, which is a small lake (4,7 Ha), little deep (maximum depth: 1.80 m; mean depth: 0.8 m), which is located in the northeast zone, specifically

in the León province (2°33′N, 5°12′W). For the details about the methodology of the experiment, see Stephen et al. (2004) and Van de Bund et al. (2004). Here we give a brief synthesis.

The experiment was carried out weekly through a period of ten weeks (9 weeks for applying treatments and one for control), in the summer of 1998 (from June 9 to August 10). For the development of the experiment, 36 mesocosms were used. Each mesocosm constitutes an experimental unit. The mesocosms were made of polyethylene (125 μm) cylinders mounted top and bottom on plastic hoops, 1 m diameter, so that the mesocosm area was 0.79 m^2 and depth about 1 m. The bottom hoop was weighted or staked so that it was firmly buried in sediment so the water in the mesocosms was in contact with the sediment and the top of the mesocosm was open to the atmosphere.

Three fish-density (0, 4 and 20 g fresh mass m^{-2}) and four nutrient-loading (N–P in mg L^{-1}: 0–0, 1–0.1, 5–0.5 and 10–1) treatments were applied, i.e., we have 12 treatments with three replications each. The treatments will be denoted as a combination of fish levels (F0, F1, F2) and nutrient levels (N0, N1, N2 , N3) in the graphs.

The fish species was *Rutilus arcasii*, a type of small fish, which is usually found in such lakes. Nutrients were provided weekly to simulate a constant external charge.

The physico-chemical variables measured were pH, total alkalinity (ALK), CO_2, total suspended solids (TSS), TP, SRP, nitrate (NO_3), ammonium (NH_4), Conductivity (CON) and Chlorophyll (CLO). One of the objectives of the experiment was to study the effects of the 12 treatments in the evolution of the set of physico-chemical variables.

At the methodological level, the objective of this work could be formulated as "investigate if treatment effects are stable over time" that is "*investigate change produced in the measurements of* 10 *variables through* 9 *occasions*, *taking into account that they were obtained for* 12 *groups* (*four nutrient levels and three fish concentration*) with three replications each". To achieve this objective it is necessary to look for a common reference system in which to interpret the change in treatments differences produced through nine weeks.

The data has been originally centered and standardized for each variable and each occasion; so the group means are estimations of the treatment effects and the calculations are based on correlations rather than covariances. Observe that proceeding in this way removes the differences between the overall means over time. Other ways of centering would be possible depending on the objectives of the researcher.

For statistical analysis using techniques as described in this work we have used programs written in MATLAB specifically for this purpose. So far they are not described in standard packages.

4.1. Intra-structure analysis

The analysis of the principal components of the matrix $\mathbf{S}^{W}$ and the graphical representation in the main plane gives the Euclidean inter-structure representation. In the Fig. 1 we have shown the graphical representation of the matrices corresponding to the nine occasions.

The percentage of inertia-variance explained for axis 1, 76.8%, is very high compared with axis 2, 8.9%. This shows that in the analysis of different times there is a common part in the first axis of inter-structure, and make sense to interpret the compromise structure. The compromise is represented by axis 1 in the figure and therefore it will be more correlated to weeks 2 and 6. In the second week the changes start showing as induced by the experimental condition whereas in the 6th week they proceeded to eliminate vegetation of the mesocosms. It seems therefore that we will make comparison only on base situations.

We also observe that first weeks appear as similar in the lower part of the plot whereas final weeks appear in the upper part.

4.2. Compromise structure and biplot induced by canonical STATIS

Fig. 2 shows the Euclidean image of the compromise. The inertia absorption in the first principal plane is 74%.

All the groups have a good quality of representation in the first plane, they are all in the interval [51.7%, 89.3%], as can be seen in Table 1. The quality of representation is interpreted as the part of the treatment variation explained by the factors and it is calculated as the squared cosine of the angle formed by each vector in the multidimensional space and its projection on the first principal plane.

On axis 1 we see the gradient on which all the nutrient quantities are placed. To the right, treatments with fewer quantities of nutrients are located whereas to the opposite side we find those with greater nutrient quantities. The axis 2 shows the difference among intermediate and extreme nutrients. The differences among fish levels are less important

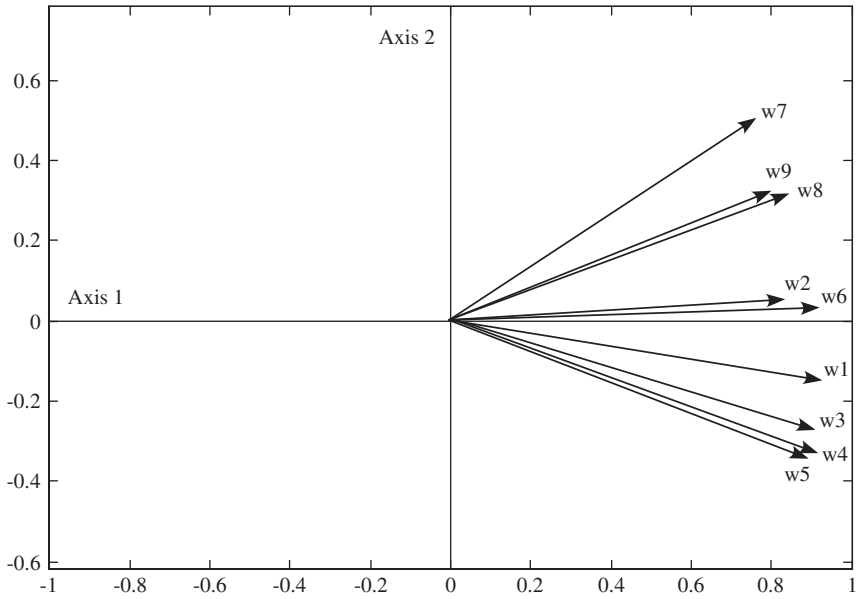


Fig. 1. Inter-structure of the 9 occasions.

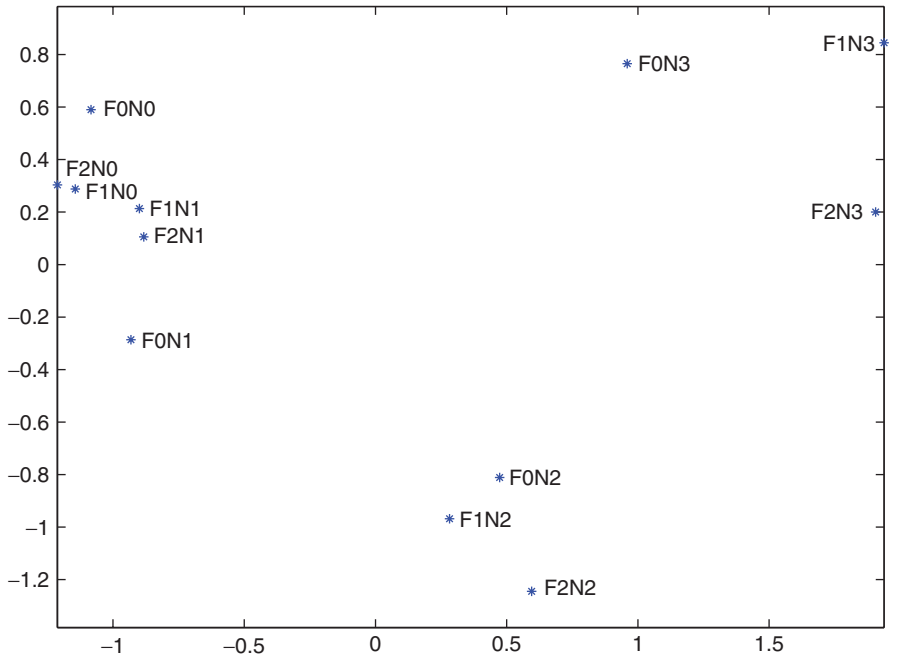


Fig. 2. Euclidean image of the compromise.

than the differences among nutrient levels. A small interaction between the two factors could be present based on the fact that the differences among fish levels are greater at the highest nutrient concentration.

Fig. 3 shows the trajectory of each treatment as obtained from the projection in (4). Although controversial, the representation helps the researcher in establishing the stability of the treatments over time. In this case there is a clear

Table 1
Quality of representation accumulated for the compromise groups over the principal axis for Canonical STATIS

Treatment	Axis1	Plane1–2
F0N0	58.24	75.51
F0N1	47.24	51.70
F0N2	*15.55*	61.32
F0N3	40.29	66.01
F1N0	69.84	74.26
F1N1	49.32	52.10
F1N2	*5.74*	73.27
F1N3	75.09	89.37
F2N0	66.48	70.64
F2N1	63.52	64.43
F2N2	*14.72*	79.21
F2N3	84.26	85.18

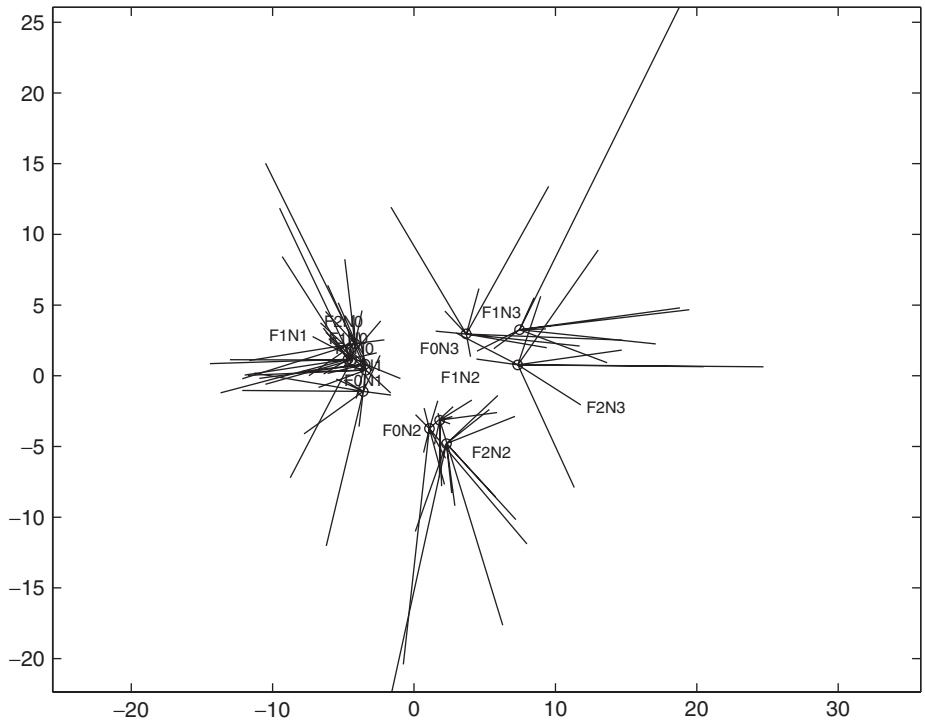


Fig. 3. Classical STATIS trajectories of the different groups.

difference, for all the weeks, between three groups formed by the first and the second levels of nutrients together and the third and fourth levels separately, reproducing the same pattern as the consensus configuration.

Table 2 shows the quality of the representation of each treatment at each week in two dimensions, calculated as the cosine ($\times 100$) of the angle between the vector in the multidimensional space and its projection onto the two-dimensional subspace. When that measure is close to 1 (100%) the variability of the treatment is accounted for the factor axis and conclusions about its behavior can be taken. If the quality of representation is close to 0 the position of the point is not interpretable. The quality of representation of he compromise/consensus mean treatments has also been added to the table. Observe that its magnitude is much higher for the consensus than for the individual weeks; in general, the quality of representation for the individual weeks is not very high.

Table 2
Quality of representation for each treatment at each occasion in the classical trajectories for STATIS

	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6
F0N0	*78.077*	*50.82*	*84.903*	*70.123*	46.698	*51.141*
F0N1	45.393	30.878	*63.427*	75.702	31.855	*74.714*
F0N2	28.559	32.805	23.214	47.210	*70.801*	28.732
F0N3	38.118	*66.297*	*67.700*	*56.643*	42.942	*50.73*
F1N0	*44.964*	*63.239*	*51.566*	*55.262*	*58.394*	*82.699*
F1N1	23.813	19.923	*67.698*	*66.208*	*69.041*	49.161
F1N2	10.407	*53.464*	26.729	18.194	12.237	48.021
F1N3	*57.963*	*72.254*	*52.279*	*84.917*	*65.148*	*61.645*
F2N0	*57.780*	27.855	34.556	71.276	35.365	62.835
F2N1	27.658	35.965	*71.755*	*61.461*	45.807	44.609
F2N2	33.790	*56.075*	*63.264*	*54.802*	43.374	36.125
F2N3	*65.939*	*81.801*	*74.530*	*62.830*	44.819	*58.254*

	Week 7	Week 8	Week 9	Consensus
F0N0	32.457	32.006	32.005	*75.51*
F0N1	13.062	4.835	16.626	*51.70*
F0N2	32.360	18.982	37.224	*61.32*
F0N3	14.971	38.525	33.264	*66.01*
F1N0	31.073	32.621	35.864	*74.26*
F1N1	1.101	6.384	35.127	*52.10*
F1N2	35.998	43.298	45.263	*73.27*
F1N3	28.087	36.312	21.976	*89.37*
F2N0	29.533	22.113	38.012	*70.64*
F2N1	19.979	16.386	41.949	*64.43*
F2N2	*50.910*	21.526	*51.738*	*79.21*
F2N3	11.669	35.021	45.091	*85.18*

The Fig. 4 shows the biplot for canonical STATIS in which the compromise means are represented by vectors, and the trajectories for the variables through the nine weeks are shown. Table 1 displays also the quality of the representation of the compromise means in the biplot.

As we had already pointed out before, mean coordinates in the picture coincide with coordinates in the biplot and are interpreted as average Mahalanobis distance among treatment means. The interpretation of the biplot is made in terms of inner products of markers in the sense that the inner product of each treatment marker by each variable marker in a specific occasion approximates the value of the treatment effect of that variable in the specific occasion. Points near the origin show that the corresponding treatment effects are all close to zero and then the means are not very different; points not near the origin show the corresponding treatment effects are different and the direction of the differences.

Table 3 shows the quality of representation for variable trajectories in the Biplot graphics. The quality of the representation of each variable is calculated as the explained sum of squares of each variable (in all occasions) in the biplot approximation. The table shows the variance explained by the first axis and the first and second together (plane 1–2).

The quality of the representation of the variables can also be calculated for each occasion as presented in Table 4. The information in Table 3 is the mean goodness of fit for all the occasions. A detailed inspection of the two tables shows that, except for some points, the quality for each week follows the same pattern as the mean.

We can see that adding the second axis does not improve much the quality of the representation, so none of the variables is clearly related to that axis. The variables with high values on the first axis are the responsible for the gradient of nutrient effects.

If all the points in the trajectory are close this indicates that the treatment effects are stable over time, in other case there are some interactions with time.

From the above table we deduce that the variables ALK, NH_4, NO_3, SRP, TP, and CON receive high contributions from the axis 1 and therefore they are decisive in differentiating among treatments; the variables PH, CO_2, TSS and

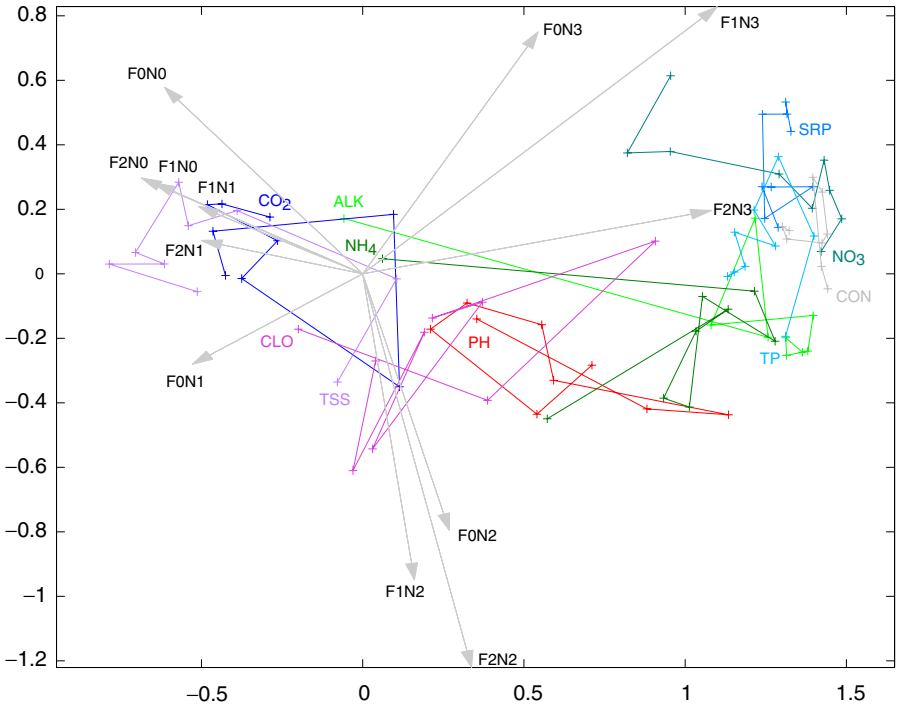


Fig. 4. Biplot for CANONICAL STATIS with the trajectories for each variable.

Table 3
Quality representation for variable trajectories in the biplot induced by the canonical STATIS

Variable	Axis 1	Plane 1–2
PH	17.77	29.31
ALK	*93.93*	*96.07*
CO_2	15.40	16.64
NH_4	*82.66*	*83.43*
NO_3	*78.85*	*95.29*
SRP	*84.73*	*88.77*
TP	*83.64*	*83.65*
TSS	32.90	32.94
CON	*94.05*	*95.24*
CLO	0.06	25.22

CLO are not well represented in this plane, so its information cannot be studied in this graph. Taking into account that this representation accounts for the most important part of the information, probably the variabilities of the effect treatments for those variables are very small compared to the others.

In Fig. 5 we have separate representations for the trajectories in Fig. 4. We see that the treatment effects for the variables NO_3, SRP, TP and CON are stable over time whit higher effects for treatment with higher amounts of nutrients. Only NO_3 trajectory show some degree of eccentricity which could indicate a gradual increment in the first weeks and then remain stable after weeks 4–5.

The variables ALK and NH_4 seem to be stable except for the first week in which the effects are close to zero. If we analyse the information week by week we notice that ALK has a very slow quality of representation in the first week (Table 4), which shows that this point has a apparent position, and the treatment differences, for this week, do not follow the general pattern. However, for the next weeks the quality of representation is very high, so we can interpret

Table 4
Quality of representation for each variable at each occasion in the biplot induced by the Canonical STATIS. Qualities higher than 50% have been highlighted in italics

Var.	W1		W2		W3	
	Ax. 1	Ax. 1–2	Eje 1	Ejes 1–2	Eje 1	Ejes 1–2
PH	35.00	40.50	*57.18*	*70.10*	*70.79*	*81.31*
ALK	0.57	5.60	*74.275*	*76.10*	*75.30*	*76.81*
CO_2	27.93	38.36	*22.65*	28.23	23.57	28.24
NH_4	0.47	0.75	*94.53*	*94.72*	*92.20*	*94.67*
NO_3	*96.82*	*97.05*	*98.03*	*99.32*	*95.41*	*98.47*
SRP	*85.10*	*94.50*	*83.38*	*97.13*	*85.65*	*97.79*
TP	*92.64*	*94.67*	*97.00*	*97.68*	*85.92*	*92.74*
TSS	1.12	21.52	1.48	1.52	19.39	24.30
CON	*98.70*	*98.80*	*93.99*	*94.01*	*90.242*	*94.39*
CLO	3.06	5.33	14.84	30.12	*81.214*	*82.22*
	W4		W5		W6	
PH	31.84	41.76	19.30	20.86	14.04	15.14
ALK	*78.02*	*79.71*	*95.74*	*96.56*	*94.25*	*97.09*
CO_2	9.58	11.02	13.12	13.14	2.48	25.89
NH_4	*70.04*	*70.35*	*69.16*	*80.68*	*77.11*	*90.27*
NO_3	*91.98*	*97.56*	*93.12*	*95.10*	*87.54*	*92.57*
SRP	*77.87*	*90.28*	*91.46*	*93.19*	*94.66*	*98.20*
TP	*88.98*	*91.34*	*95.09*	*95.52*	*80.25*	*81.25*
TSS	27.21	29.28	21.63	26.98	36.80	37.13
CON	*94.18*	*97.16*	*97.35*	*98.06*	*97.31*	*97.75*
CLO	4.86	6.83	11.17	11.80	0.071	22.99
	W7		W8		W9	
PH	4.04	6.74	17.77	29.31	17.77	29.31
ALK	*92.77*	*96.21*	*93.93*	*96.07*	*93.93*	*96.07*
CO_2	1.27	6.02	15.40	16.64	15.40	16.64
NH_4	*84.11*	*86.60*	*82.66*	*83.43*	*82.66*	*83.43*
NO_3	*68.68*	*79.53*	*78.85*	*95.29*	*78.85*	*95.29*
SRP	*90.73*	*94.82*	*84.73*	*88.77*	*84.73*	88.77
TP	*90.02*	*90.05*	*83.64*	*83.65*	*83.64*	*83.65*
TSS	25.42	25.48	32.90	32.94	32.90	32.94
CON	*94.02*	*94.66*	*94.05*	*95.24*	*94.05*	*95.24*
CLO	5.29	10.06	0.06	25.22	0.06	25.22

these points of the trajectory without problem. This situation produces a jump between the points that represent the week 1 and the week 2 that may be important, the increase in their values is not clear after two weeks of treatment.

For the rest of the variables with high quality, its increment in the mesocosms that received higher amounts of nutrients is evident from the first week of the study.

Finally observe that the magnitude of the goodness of fit (quality of representations) in Table 4 is much higher than the magnitude of the qualities in Table 2, showing that the trajectories induced by the biplot, have a better fit than the classical trajectories.

5. Conclusions

We have proposed an alternative method to the classical method STATIS-ACT that excels its limitations when the experimental units are classified in groups, for example, when the data are coming from the application of treatments.

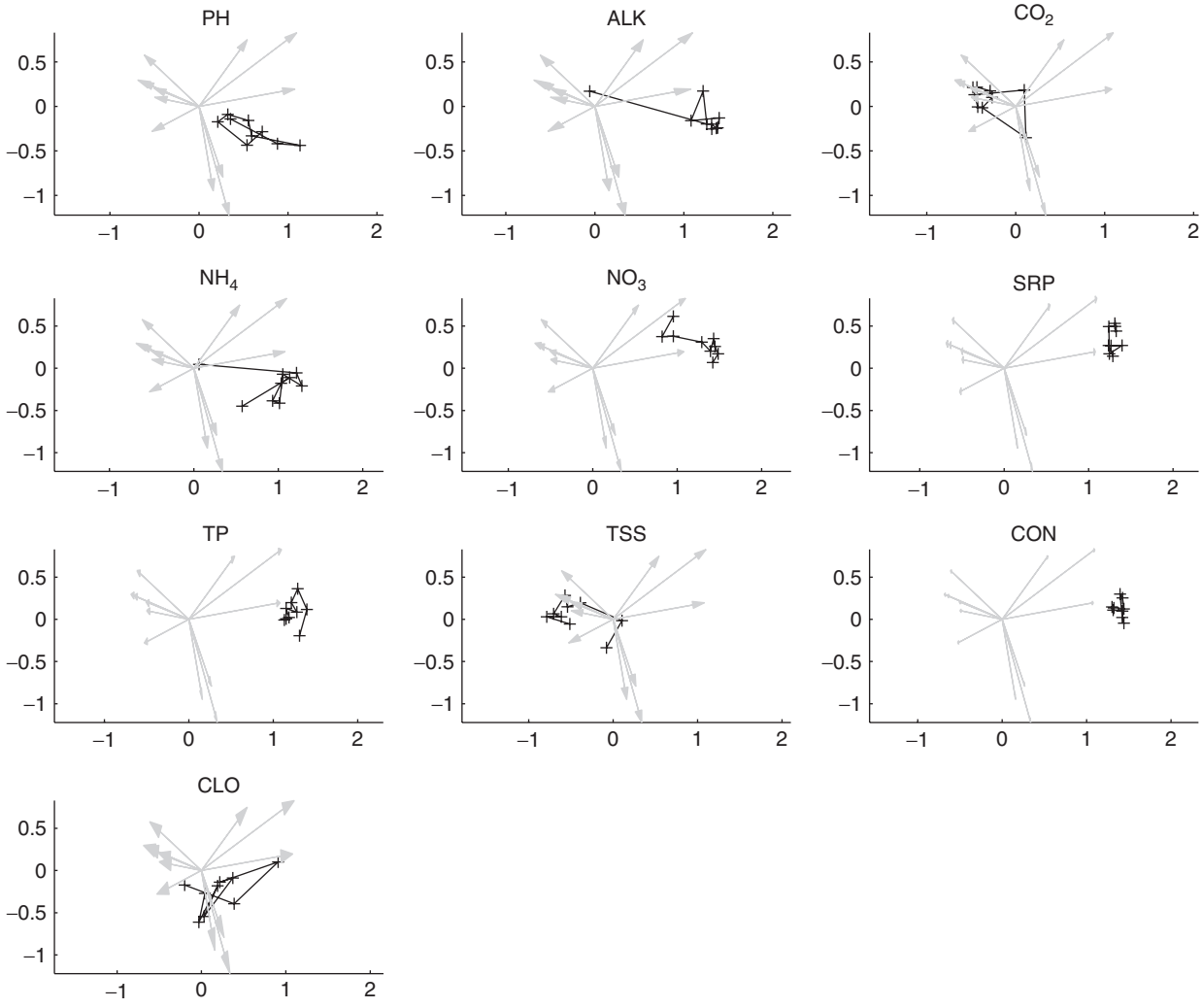


Fig. 5. Separate trajectories for each variable.

In this paper, we proposed a technique called canonical STATIS because is an extension of the canonical variate analysis using STATIS-ACT methodology.

Canonical STATIS uses the Mahalanobis distances instead of the Euclidean distances and extracts the stable part, along occasions, of the behavior of the groups by obtaining a consensus structure and a common reference system.

The Biplot representation induced by Canonical STATIS permits the projection of the matrices of means (or effects) of the treatments on the compromise structure. With these projections we can get a common reference system for all the variables in all the occasions, with a clear interpretation, and with a quality of representation higher than the classical STATIS-ACT trajectories.

Finally, by applying the method proposed in this article to the data of European project *Shallow Wetland Lake Function and Restoration in a Changing European Environment*, we have shown the practical application of this proposal.

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